

1st Normal Form

1. Every column/ attribute needs to have a single value
2. Each row should be unique. Either through single or multiple columns. Not mandatory to have a primary key

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	1NF (First Normal Form)												
2	1. Every column/attribute need to have a single value.												
3	2. Each row should be unique. Either through a single or multiple columns. Not mandatory to have primary key												
4													
5	ORDER_NUMBER	QUANTITY	ITEM_PRICE	TOTAL_COST	ORDER_DATE	STATUS	PRODUCT_CODE	PRODUCT_NAME	PRODUCT_PRICE	CUSTOMER_NAME	PHONE	ADDRESSLINE1	ADDRESSLINE2
6	10101	30	202	6060	2003-03-24	Shipped	PRD_1001	Motorcycles	200	Land of Toys Inc.	2125557818	897 Long Airport Avenue	
7	10102	34	550	18700	2003-07-05	Cancelled	PRD_1002	Vintage Cars	500	Reims Collectables	2123356666	59 rue de l'Abbaye	
8	10103	41	420	17220	2003-01-07	Shipped	PRD_1003	Classic Cars	400	Lyon Souvenirs	2121257800	27 rue du Colonel Pierre Avia	
9	10104	45	780	35100	2003-08-25	On Hold	PRD_1004	Buses	800	Toys4GrownUps.com	6265557265	78934 Hillside Dr.	
10	10105	49	850	41650	2003-10-10	Shipped	PRD_1005	Trucks	900	Corporate Gift Ideas Co.	6505551386	7734 Strong St.	
11	10106	4	780	3120	2005-05-07	Shipped	PRD_1004	Buses	800	Auto Canal Petit	6577771000	25, rue Lauriston	
12	10106	4	850	3400	2005-05-07	Shipped	PRD_1005	Trucks	900	Auto Canal Petit	6577771000	25, rue Lauriston	
13													
14													

This table is in 1nf

2nd Normal Form

1. Must be in 1nf
2. All non-key attributes must be fully dependent on the candidate key. If a non-key attribute is partially dependent on a candidate key then separate them into different tables
3. Every table should have a primary key and the relationship between tables should be established by using a foreign key

The primary key after the 1nf form is the combination of the order number and product code. But not all columns are fully dependent on it. Some columns are only dependent on order number and some are dependent on product code. So we will split the tables into 3 parts.

The orders table

The products table

And the Customer table.

Now we will follow the next steps

ORDERS									
ORDER_NUMBER	QUANTITY	ITEM_PRICE	TOTAL_COST	ORDER_DATE	STATUS	SALES_CONTACT_LASTNAME	SALES_CONTACT_FIRSTNAME	DEAL_SIZE	
10101	30	202	6060	2003-03-24	Shipped	Yu	Kwai	Small	
10102	34	550	18700	2003-07-05	Cancelled	Henriot	Paul	Medium	
10103	41	420	17220	2003-01-07	Shipped	Da Cunha	Daniel	Medium	
10104	45	780	35100	2003-08-25	On Hold	Young	Julie	High	
10105	49	850	41650	2003-10-10	Shipped	Brown	Julie	High	
10106	4	780	3120	2005-05-07	Shipped	Perrier	Dominique	Small	
10106	4	850	3400	2005-05-07	Shipped	Perrier	Dominique	Small	

Candidate key = Order_number || Product_code

PRODUCTS		
PRODUCT_CODE	PRODUCT_NAME	PRODUCT_PRICE
PRD_1001	Motorcycles	200
PRD_1002	Vintage Cars	500
PRD_1003	Classic Cars	400
PRD_1004	Buses	800
PRD_1005	Trucks	900
PRD_1004	Buses	800
PRD_1005	Trucks	900

CUSTOMERS									
CUSTOMER_NAME	PHONE	ADDRESSLINE1	ADDRESSLINE2	CITY	STATE	POSTALCODE	COUNTRY	TERRITORY	
Land of Toys Inc.	2125557818	897 Long Airport Avenue		NYC	NY	10022	USA	NA	
Reims Collectables	2123356666	59 rue de l'Abbaye		Reims		51100	France	EMEA	
Lyon Souveniers	2121257800	27 rue du Colonel Pierre Avia		Paris		75508	France	EMEA	
Toys4GrownUps.com	6265557265	78934 Hillside Dr.		Pasadena	CA	90003	USA	NA	
Corporate Gift Ideas Co.	6505551386	7734 Strong St.		San Francisco	CA		USA	NA	
Auto Canal Petit	6577771000	25, rue Lauriston		Paris		75016	France	EMEA	
Auto Canal Petit	6577771000	25, rue Lauriston		Paris		75016	France	EMEA	

1. The table with order numbers. But there are duplicate order numbers so we cannot treat the order number as a primary key. Instead, we create a different primary key for the table which we will call order_ID

ORDERS									
ORDER_ID	ORDER_NUMBER	QUANTITY	ITEM_PRICE	TOTAL_COST	ORDER_DATE	STATUS	SALES_CONTACT_LASTNAME	SALES_CONTACT_FIRSTNAME	DEAL_SIZE
1	10101	30	202	6060	2003-03-24	Shipped	Yu	Kwai	Small
2	10102	34	550	18700	2003-07-05	Cancelled	Henriot	Paul	Medium
3	10103	41	420	17220	2003-01-07	Shipped	Da Cunha	Daniel	Medium
4	10104	45	780	35100	2003-08-25	On Hold	Young	Julie	High
5	10105	49	850	41650	2003-10-10	Shipped	Brown	Julie	High
6	10106	4	780	3120	2005-05-07	Shipped	Perrier	Dominique	Small
7	10106	4	850	3400	2005-05-07	Shipped	Perrier	Dominique	Small

2. For the products table, there are whole duplicate rows. So we will remove them and then treat the Product_Code as the primary key

PRODUCTS		
PRODUCT_CODE	PRODUCT_NAME	PRODUCT_PRICE
PRD_1001	Motorcycles	200
PRD_1002	Vintage Cars	500
PRD_1003	Classic Cars	400
PRD_1004	Buses	800
PRD_1005	Trucks	900

3. Then comes the customers' table. We do not have a primary key here. First, we will remove duplicate records here too. Then we will create a new column names Customer_ID then declare it as primary key.

CUSTOMERS									
CUSTOMER_ID	CUSTOMER_NAME	PHONE	ADDRESSLINE1	ADDRESSLINE2	CITY	STATE	POSTALCODE	COUNTRY	TERRITORY
C1	Land of Toys Inc.	2125557818	897 Long Airport Avenue		NYC	NY	10022	USA	NA
C2	Reims Collectables	2123356666	59 rue de l'Abbaye		Reims		51100	France	EMEA
C3	Lyon Souveniers	2121257800	27 rue du Colonel Pierre Avia		Paris		75508	France	EMEA
C4	Toys4GrownUps.com	6265557265	78934 Hillside Dr.		Pasadena	CA	90003	USA	NA
C5	Corporate Gift Ideas Co.	6505551386	7734 Strong St.		San Francisco	CA		USA	NA
C6	Auto Canal Petit	6577771000	25, rue Lauriston		Paris		75016	France	EMEA

- The next thing to do is build relationships between the 3 tables. We can do it by adding foreign keys to the existing tables or we can just create a new table that establishes the relationships.

RELATIONSHIPS		
ORDER_ID	PRODUCT_CODE	CUSTOMER_ID
1	PRD_1001	C1
2	PRD_1002	C2
3	PRD_1003	C3
4	PRD_1004	C4
5	PRD_1005	C5
6	PRD_1004	C6
7	PRD_1005	C6

The 2nd normal form is now complete.

3rd Normal Form

- Must be in 2nf
- Avoid transitive dependencies

Transitive Dependency:

Lets say you have a table T which has 3 columns namely A, B and C.

If A is functionally dependent on B, and B is functionally dependent on C then we can say that A is functionally dependent on C.

In the orders table, if we know the name of the customer, we can find out the order he purchased. Then if we know the order, we can find out the name of the salesperson which is Sales_person_firstname and Sales_person_LastName combined. So let's just separate those two columns into a different new table named Employees. Then we will add it to the relationship table by removing duplicate rows and creating a primary key named Emp_ID

EMP_ID	SALES_CONTACT_LASTNAME	SALES_CONTACT_FIRSTNAME
E1	Yu +	Kwai
E2	Henriot	Paul
E3	Da Cunha	Daniel
E4	Young	Julie
E5	Brown	Julie
E6	Perrier	Dominique

RELATIONSHIPS			
ORDER_ID	PRODUCT_CODE	CUSTOMER_ID	EMP_ID
1	PRD_1001	C1	E1
2	PRD_1002	C2	E2
3	PRD_1003	C3	E3
4	PRD_1004	C4	E4
5	PRD_1005	C5	E5
6	PRD_1004	C6	E6
7	PRD_1005	C6	E6

The database is now in the 3rd normal form.

Boyce Codd Normal Form(BCNF)

A relation schema R is in BCNF with respect to a set F of functional dependencies if for all functional dependencies in F^+ of the form

$$\alpha \rightarrow \beta$$

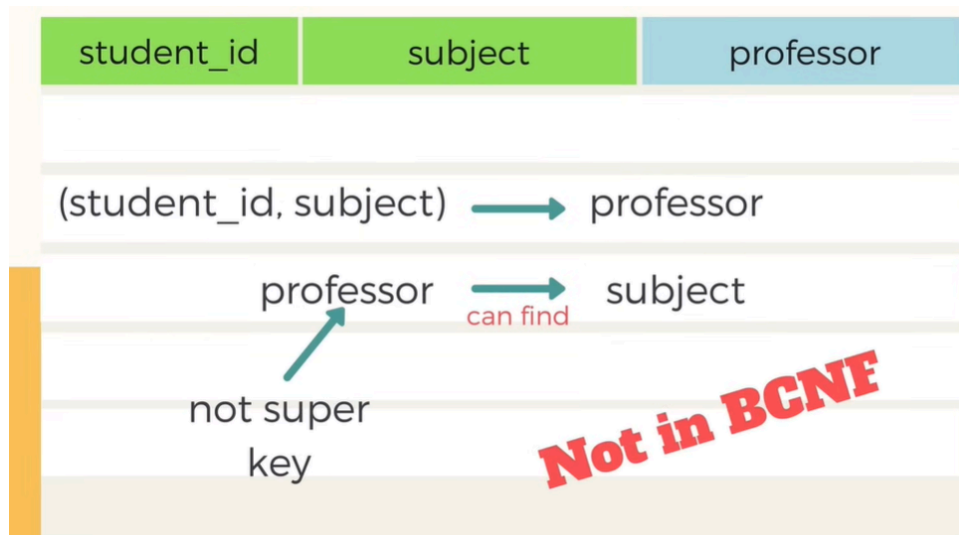
where $\alpha \subseteq R$ and $\beta \subseteq R$, at least one of the following holds:

- $\alpha \rightarrow \beta$ is trivial (i.e., $\beta \subseteq \alpha$)
- α is a superkey for R

- It should be in the 3rd Normal Form.
- For any dependency $A \rightarrow B$, A should be a **super key**.



Here, the student id and subject form the primary key combined. The together determine the rest of the table. But, the professor is an non key attribute and it can determine the subject which is a non key attribute. This violates BCNF.



To solve this, we have to break the table into two parts—one containing the student id and the professor ID. Other contains professor id, professor, and subject.

